What is an RS (ReplicaSet) in Kubernetes?

A **ReplicaSet (RS)** is a Kubernetes controller that ensures a specified number of **identical pod replicas** are running at all times.

ReplicaSet is the **next-generation replacement** for ReplicationController and is used by **Deployments** behind the scenes.

Key Features of ReplicaSet:

- Ensures **self-healing** of pods (recreates them if they crash or get deleted).
- Uses **label selectors** to manage pods.
- Commonly managed by **Deployments** (direct use of RS is rare).

Example: ReplicaSet YAML

```
apiVersion: apps/v1
kind: ReplicaSet
metadata:
 name: nginx-rs
spec:
 replicas: 3
 selector:
  matchLabels:
   app: nginx
 template:
  metadata:
   labels:
    app: nginx
  spec:
   containers:
    - name: nginx-container
      image: nginx:latest
      ports:
      - containerPort: 80
```

Explanation:

- **replicas**: Ensures 3 pod replicas are always running.
- **selector.matchLabels**: Matches the pods created using the template.
- **template**: Describes the pod (container name, image, port, etc.).

How to Create the ReplicaSet:

kubectl apply -f nginx-rs.yaml

View the RS and Pods:

kubectl get rs

kubectl get pods

Difference Between RC and RS:

Feature	RC	RS
API Group	v1	apps/v1
Label Selector Type	Only equality-based	Supports set-based selectors
Common Use	Deprecated	Used with Deployments